



Initial Security Hardening, Threat Tracking & Compliance Auditing

A hands-on AWS Academy lab guiding you through building a secure cloud environment, activating intelligent threat detection, and auditing your infrastructure against industry compliance frameworks – all within your sandboxed Learner Lab environment.

AWS ACADEMY LAB

ESTIMATED TIME: 75 MINUTES

REGION: US-EAST-1

Lab Overview & Learning Objectives

In this lab, you will build a foundational security architecture from scratch, then activate and validate three core AWS security services. By the end, you will have hands-on experience with GuardDuty, Security Hub, and Trusted Advisor working in concert.

01

Prepare Your Sandbox

Initialize your AWS Academy Learner Lab and verify region alignment to us-east-1.

03

Activate Security Services

Enable GuardDuty for threat detection and Security Hub for centralized posture management.

02

Build Core Infrastructure

Construct a VPC, subnet, internet gateway, route table, and security group from scratch.

04

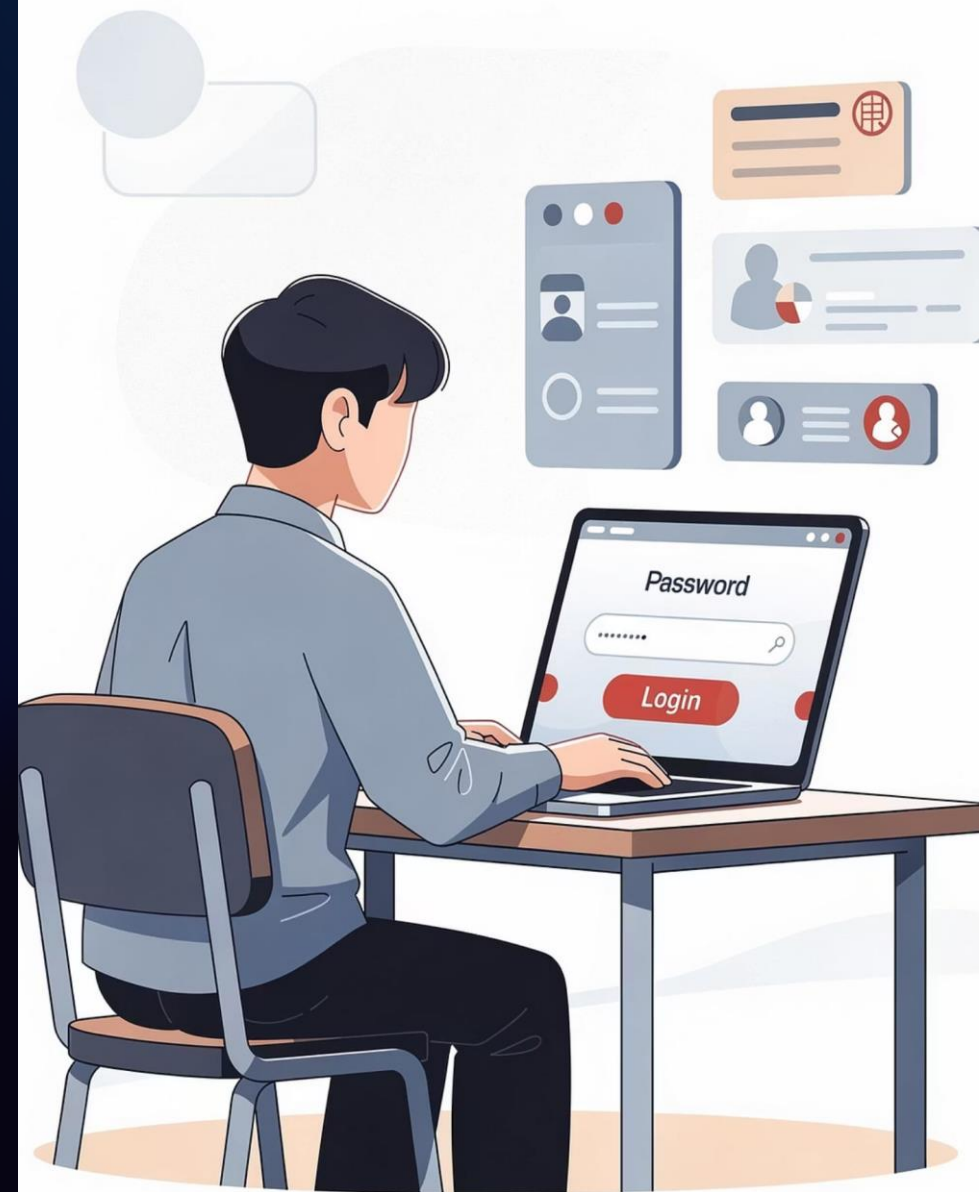
Audit & Clean Up

Run Trusted Advisor compliance checks, validate findings, and tear down all lab resources.

SECTION 1

Prerequisites & Sandbox Preparation

Before you write a single line of configuration, your sandbox environment must be fully active and correctly targeted. Follow these steps precisely before proceeding to any infrastructure build tasks.



Starting Your Learner Lab

1 Log In to AWS Academy Portal

Use your student credentials to access your designated AWS Academy Portal and navigate to your course shell.

2 Launch the Lab Environment

Click the **Learner Lab** link, then click the **Start Lab** (play icon) button to provision your isolated container environment.

3 Wait for Green Status Indicator

Monitor the status circle next to the AWS console link. Do not proceed until it turns from **amber to solid green** – this confirms full provisioning.

4 Verify Region: N. Virginia (us-east-1)

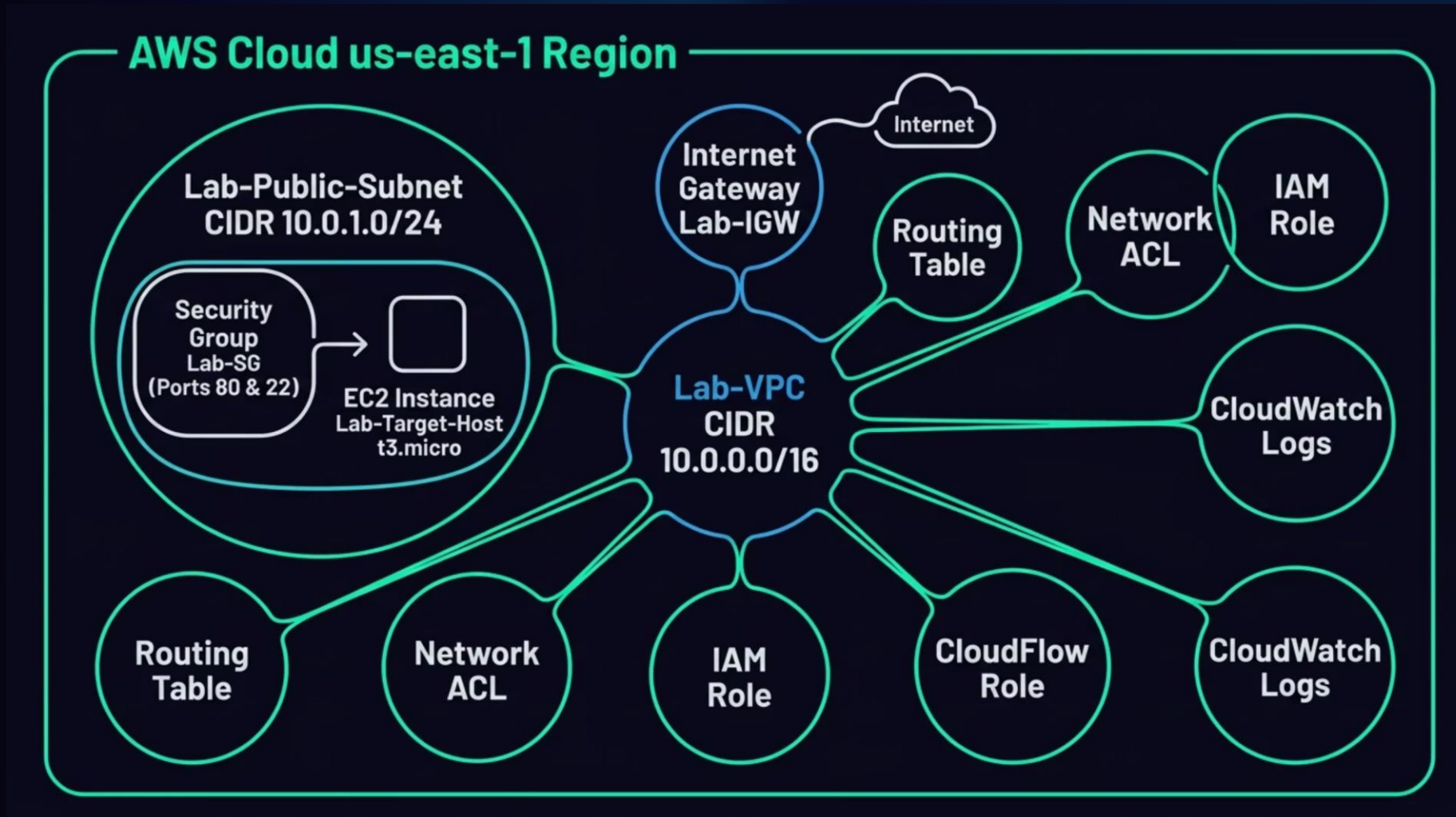
After the console opens, check the top-right dropdown. If the region is not **us-east-1**, click the dropdown and re-select **N. Virginia** before continuing.



Critical Checkpoint: Never proceed past this step until the status indicator is solid green AND your region reads us-east-1. Misalignment causes resource provisioning failures downstream.

Core Infrastructure Assembly

To monitor threat signals accurately, we must first build a basic but complete network environment from scratch. The diagram below shows the architecture you will construct in this section.



Step 1 — Create the VPC & Subnet

Create Lab-VPC

1. Search **VPC** in the top global search bar and select the service.
2. In the left pane, choose **Your VPCs**, then click **Create VPC**.
3. Set **Resources to create** → **VPC only**.
4. Set **Name tag** → **Lab-VPC**.
5. Set **IPv4 CIDR** → **10.0.0.0/16** (manual input).
6. Click **Create VPC**.

Create Lab-Public-Subnet

1. In the left pane, choose **Subnets**, then click **Create subnet**.
2. For **VPC ID**, select your newly created **Lab-VPC**.
3. Set **Subnet name** → **Lab-Public-Subnet**.
4. Set **IPv4 CIDR block** → **10.0.1.0/24**.
5. Click **Create subnet**.

Step 2 — Internet Gateway & Route Table

Create & Attach Lab-IGW

1. In the left pane, choose **Internet gateways** → **Create internet gateway**.
2. Set **Name tag** → **Lab-IGW**. Click **Create internet gateway**.
3. With the gateway selected, click **Actions** → **Attach to VPC**.
4. Select **Lab-VPC** and click **Attach internet gateway**.

Configure the Route Table

1. In the left pane, choose **Route tables**. Select the main route table for **Lab-VPC**.
2. Click the **Routes** tab in the bottom panel → **Edit routes**.
3. Click **Add route**. Set **Destination** → **0.0.0.0/0**.
4. Set **Target** → **Internet Gateway** → select **Lab-IGW**.
5. Click **Save changes**.

Step 3 — Create the Security Group (Lab-SG)

A security group acts as a virtual firewall for your EC2 instance. For this lab, you will open Port 22 (SSH) and Port 80 (HTTP) — both of which Trusted Advisor will later flag as intentional exposure risks to validate.

Basic Configuration

- **Security group name:** Lab-SG
- **Description:** Security group for local lab evaluation
- **VPC:** Remove the default and select **Lab-VPC**

Inbound Rule 1 — SSH

- **Type:** SSH
- **Port:** 22 (auto-populated)
- **Source:** Anywhere-IPv4 (0.0.0.0/0)

Inbound Rule 2 — HTTP

- **Type:** HTTP
- **Port:** 80 (auto-populated)
- **Source:** Anywhere-IPv4 (0.0.0.0/0)

Click **Create security group** at the bottom of the form to finalize.



SECTION 3

Provisioning the Target Compute Node

With your network scaffolding in place, you will now launch a cost-efficient EC2 virtual instance inside your custom VPC to serve as the active infrastructure target for your security monitoring tools.

EC2 Launch Configuration

Search **EC2** in the top global search bar, open the dashboard, and click **Launch instance**. Apply the following configuration precisely.

Instance Identity

- **Name:** Lab-Target-Host
- **AMI:** Amazon Linux 2023 (default)
- **Instance Type:** t3.micro (or t2.micro if unavailable)
- **Key pair:** Proceed without a key pair

Network Settings

- **VPC:** Lab-VPC
- **Subnet:** Lab-Public-Subnet
- **Auto-assign public IP:** Enable
- **Security group:** Select existing → Lab-SG

Scroll to the bottom and click **Launch instance**. Wait for the success confirmation banner before proceeding to the next section.



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SECTION 4

Activating the Threat Intelligence Engine

Amazon GuardDuty

GuardDuty is AWS's intelligent continuous threat detection service. It analyzes CloudTrail event logs, VPC Flow Logs, and DNS logs out-of-band – without requiring you to deploy any agents – to identify malicious activity and unauthorized behavior.

Enabling GuardDuty & Security Hub

Enable Amazon GuardDuty

1. Search **GuardDuty** in the top bar and select the service.
2. On the landing page, click **Get started**.
3. Review the service-linked role permissions summary — GuardDuty needs read access to your logs.
4. Click **Enable GuardDuty**.
5. **Validation Task 1:** Confirm the left navigation pane shows **Status: Running**. Click **Findings** — the panel may be empty in a fresh sandbox. Leave this open.

Enable AWS Security Hub

1. Search **Security Hub** in the top bar and select the service.
2. Click **Go to Security Hub**.
3. Review the compliance framework toggles (AWS Foundational Security Best Practices, CIS AWS Foundations Benchmark). Leave defaults as-is.
4. Click **Enable Security Hub**.
5. **Validation Task 2:** Navigate to **Insights** in the left pane. Verify filters exist for "Top findings by resource" and similar categories.

-  Security Hub acts as your centralized security operations control plane — aggregating findings from GuardDuty and other services into a single prioritized dashboard view.

Performing Compliance Audits with AWS Trusted Advisor

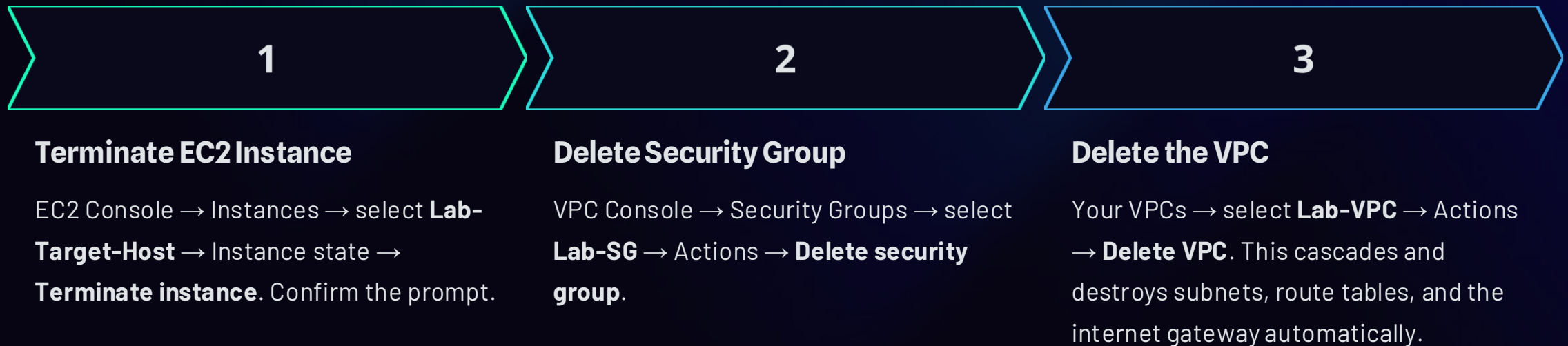
Trusted Advisor performs a holistic diagnostic scan of your cloud account across five foundational pillars. Navigate to **Trusted Advisor** via the top search bar and click **Dashboard** in the left navigation panel.

Evaluation Pillar	Status	Definition
Cost Optimization	✔ Green	Resource expenditures align with recommended baseline patterns.
Performance	✔ Green	Computing architecture parameters are balanced for throughput.
Security	⚠ Orange / ● Red	Account configurations exhibit exposure risks requiring validation review.
Fault Tolerance	🔔 Investigation	Redundancy structures require secondary AZ load balancing additions.
Service Limits	✔ Green	Infrastructure usage falls well within allocated cloud thresholds.

Validation Task 3: Click the **Security** tab and expand **Security Groups — Specific Ports Unrestricted**. You will see **Lab-SG** flagged for exposing Port 22 and Port 80 to 0.0.0.0/0. This confirms Trusted Advisor is actively scanning your architecture in real time.

Mandatory Cleanup: Tear Down All Lab Resources

To protect your **\$100 allocated lab budget** and keep your sandbox clean for future labs, you must terminate every resource you created – in the correct sequence – immediately after completing all validations.



 Do not skip cleanup. Leaving running EC2 instances or NAT Gateways active will rapidly consume your lab credit balance, which cannot be restored.

SECTION 8

Troubleshooting Common Lab Failures

If you encounter errors during any stage of this lab, consult the matrix below before requesting instructor assistance. Most failures have a fast, self-service resolution.

Observed Symptom	Root Cause	Remediation Steps
"Access Denied" or "Unauthorized" error banner appears.	Attempting to activate paid/premium tiers blocked by AWS Academy SCP policies.	Stop immediately. Verify you selected VPC only and standard defaults. Do not attempt to enable premium features.
GuardDuty enable step fails with a service role dependency error.	Learner container IAM permissions have not fully synchronized yet.	Wait 60 seconds, refresh the browser tab to re-read active IAM identities, then click Enable again.
EC2 launch wizard states the target AMI is unavailable.	The active region has drifted away from the required us-east-1 zone.	Check the top-right header dropdown. If it is not N. Virginia (us-east-1) , switch back and recreate your launch settings.